

Three-Dimensional View Relationship-Based Context-Aware Emotion Recognition

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Abstract—Context-aware emotion recognition (CAER) leverages comprehensive scene information, including facial expressions, body postures, and contextual background. However, current studies predominantly rely on facial expressions, body postures, and global contextual features; the interaction between the agents (target individuals) and other objects in the scene is usually absent or incomplete. In this article, a three-dimensional view relationship-based CAER (TDRCer) method is proposed, which comprises two branches: the personal emotional branch (PEB) and the contextual emotional branch (CEB). First, PEB is designed for the extraction of facial expression features and body posture features from the agent. A vision transformer (ViT), pretrained by contrastive learning with a novel loss function combining Euclidean distance and cosine similarity, is applied to enhance the robustness of facial expression features. Meanwhile, the human body contour images extracted by semantic segmentation are fed into another ViT to extract body posture features. Second, CEB is constructed for the extraction of global contextual features and interactive relationships among objects in the scene. The images masked by the agents' bodies are fed into a ViT to extract global contextual features. By leveraging both the gaze angle and depth map, a three-dimensional view graph (3DVG) is constructed to represent the interactive relationships between agents and objects in the scene. Then, a graph convolutional network is employed to extract interactive relationship features from the 3DVG. Finally, the multiplicative fusion strategy is applied to fuse the features of two branches, and the fused features are utilized to classify the emotions. TDRCer achieves an accuracy of 89.90% on the CAER-S dataset and a mean average precision (mAP) of 36.02% on the EMOTIONS in context (EMOTIC) dataset. The code can be accessed at <https://github.com/mengTender/TDRCer>.

Index Terms—Context-aware emotion recognition (CAER), graph convolutional network, interactive relationship, three-dimensional view relationship representation, vision transformer (ViT).

Received 27 October 2022; revised 12 December 2023 and 4 September 2024; accepted 30 September 2024. Date of publication 21 October 2024; date of current version 9 July 2025. This work was supported in part by the Key Research and Development Program of Shandong Province, China, under Grant 2021SFGC0104; in part by the Key Research and Development Program of Shandong Province, China (Major Scientific and Technological Innovation Project), under Grant 2021CXGC010506; in part by the Natural Science Foundation of Shandong Province, China, under Grant ZR2022LZH003, Grant ZR2021MF118, and Grant ZR2020LZH008; and in part by the National Natural Science Foundation of China under Grant 62101311. (*Corresponding author: Xiangwei Zheng.*)

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Digital Object Identifier 10.1109/TNNLS.2024.3476249

I. INTRODUCTION

RECENTLY, emotion recognition has been widely utilized in various fields, such as telemedicine [1], fatigue monitoring [2], and human-computer interaction [3]. There are various emotion models, and among them, the discrete emotion (*Happy, Sad, Disgust, Angry, Surprise, Neutral, and Fear*) model and the *Valence, Arousal, and Dominance* (VAD) model are frequently utilized [4]. Human emotions can be expressed in various ways, such as facial expressions [5], [6], [7], [8], [9], text [10], [11], speech [12], [13], and physiological signals. Most early emotion recognition studies were based on one or more of the aforementioned emotional cues [14], [15], [16], [17], [18]. Among the various emotional cues, facial expressions are deemed the most direct and natural means of conveying emotions. Conventional emotion recognition relying on visual cues primarily focuses on facial expressions. However, there are some drawbacks to exclusively depending on facial expressions. On the one hand, facial information might be obscured or blurred in expansive scenes, posing challenges for emotion recognition methods [19]. On the other hand, due to inherent similarities in facial appearance and muscle movements between distinct emotions, such as *Happy* and *Surprise*, the exclusive reliance on facial expressions might lead to confusion among different emotional categories.

In emotion recognition, researchers have explored the extraction of additional information from images to effectively represent emotional states. These studies mainly focus on scene context and body posture [20], [21], [22] of agents (target individuals). The reflection of the scene on emotions primarily concentrates on two aspects: the overall scene context and the interaction between the agents and surrounding objects [23]. Apart from facial expressions, body posture represents another significant means by which agents convey their emotions to the external world [24]. The significance of scene context and body posture in emotion recognition has already been explored in several works [25], [26], [27], [28], which are referred to as context-aware emotion recognition (CAER). In fact, the entire images are frequently employed to represent the scene context of the agents. However, the absence of interaction between the agents and other objects in the scene leads to diminished accuracy in emotion recognition.

To address the challenges of the incompleteness of facial information and the absence of interactive features between agents and surrounding objects, we propose a three-dimensional view relationship-based CAER (TDRCer) method. There are two branches in TDRCer, namely, the personal emotional branch (PEB) and the contextual emotional

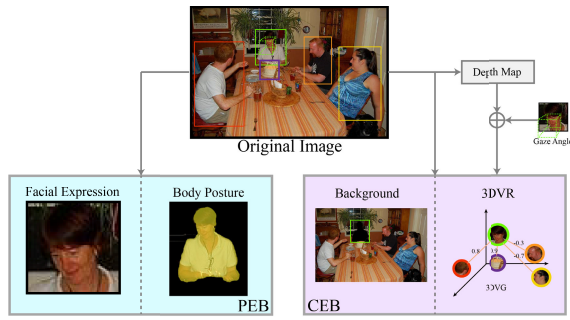


Fig. 1. Proposed method consists of two branches: PEB and CEB. PEB is employed for extracting personal features, encompassing facial expressions and body postures. CEB is utilized to extract scene features, including global context features and the interactive relationship between agents and surrounding objects.

branch (CEB), extracting personal emotional features and contextual emotional features, respectively. An illustration of the proposed method is shown in Fig. 1. In PEB, the features of facial expression and body posture are extracted and fused, and two vision transformers (ViTs) [30] are introduced as the backbones. Contrastive learning is applied to pretrain a ViT for the robust feature extraction of facial expressions. In CEB, the features of global scene context (background) and the features of interactive relationships among objects are extracted. A three-dimensional view graph (3DVG) is constructed with the gaze angle and depth map to effectively represent the interaction between agents and surrounding objects. The multiplicative fusion strategy is finally applied to fuse the features extracted by PEB and CEB.

The main contributions of this article are given as follows.

- 1) We propose a TDRCer method that effectively leverages the features of facial expression, body postures, global scene context, and the information of interactive relationships between agents and surrounding objects.
- 2) We employ a contrastive learning approach to pretrain the ViT in PEB and propose a novel loss function that combines Euclidean distance and cosine similarity as the optimization objective for pretraining.
- 3) We construct a 3DVG by incorporating gaze angle and depth maps, which can effectively represent the interactive relationship between agents and surrounding objects in CEB.

The rest of this article is organized as follows. Section II provides a review of the related work. Section III introduces the proposed TDRCer method, which effectively utilizes both personal and contextual scene emotional features. Section IV presents the datasets used for evaluation and discusses the experiment results by comparing them with other methods. Section V concludes the proposed method and indicates the future work.

II. RELATED WORK

A. Emotion Recognition Based on Facial Expression and Body Posture

Psychologist Mehrabian's research [31] revealed that emotional expression comprises three components: 7% speech, 38% voice, and 55% facial expression. Numerous studies have been conducted on facial expression recognition [4], [7].

Meanwhile, psychological studies indicated that body posture can effectively convey emotional information as another important mode of human communication. Pollick et al. [24] have shown that an individual's emotions can be more accurately discriminated with the help of specific arm movements. Researchers in emotion recognition have also shown considerable interest in human pose-based affective computing. Yin et al. [32] proposed MSA-GCN, which applied the spatial-temporal of gait to identify the emotion category. Yan et al. [33] presented a spatiotemporal graph-based model for human pose feature extraction, which utilized a frame sampling technique to extract image frames from videos, identified key points of the human skeleton for each frame, and established a graph structure for emotion recognition. However, this method struggles to capture long-term dependencies between actions and faces challenges in acquiring global information. To address this problem, Cho et al. [34] presented a solution that employed the transformer to capture long-range dependencies. Saha et al. [35] utilized geometric features from key points of the human skeleton to recognize gestures associated with fundamental human emotions. In the domain of affective computing by multibranch fusion, Kosti et al. [17] introduced a decision fusion model for emotion recognition tasks, leveraging high-level features from facial action units and human gestures. Gunes and Piccardi [15] proposed a dual-model approach based on body posture features and facial expression features; they employed two convolutional neural networks (CNNs) to extract body posture features and facial expression features independently and fused by feature level and decision level for emotion recognition.

However, emotion recognition relying on facial expressions and body language often neglects the impact of the surroundings on individuals' emotions. Such absence of the features from surroundings may result in the confusion of different emotion categories, thereby diminishing the accuracy of emotion recognition.

B. Context-Aware Emotion Recognition

The surroundings and interactive relationship between agents and objects also contribute to the reflection of their emotional states in the wild. To acquire comprehensive emotional information, researchers explored the methods for capturing emotional cues present in the scene context [36], [37]. Liu and Wang [14] proposed a two-branch network incorporating an attention mechanism to extract facial features and contextual features for emotion recognition. Li et al. [39] proposed SIB-Net that utilized facial expression, body posture, and global context information for emotion recognition. Ruan et al. [40] proposed a context-aware generative network that integrates scene and body information, the attention mechanism identifies key locations of emotional expressions in the video, and the label embedding layer fuses global and local features. Earlier methods extracted scene context features from the whole images without representing the interaction among the objects in the scene. As the research advanced, efforts were initiated to model the emotional information manifested by the interaction of objects in the scene. Ledgerwood [41]

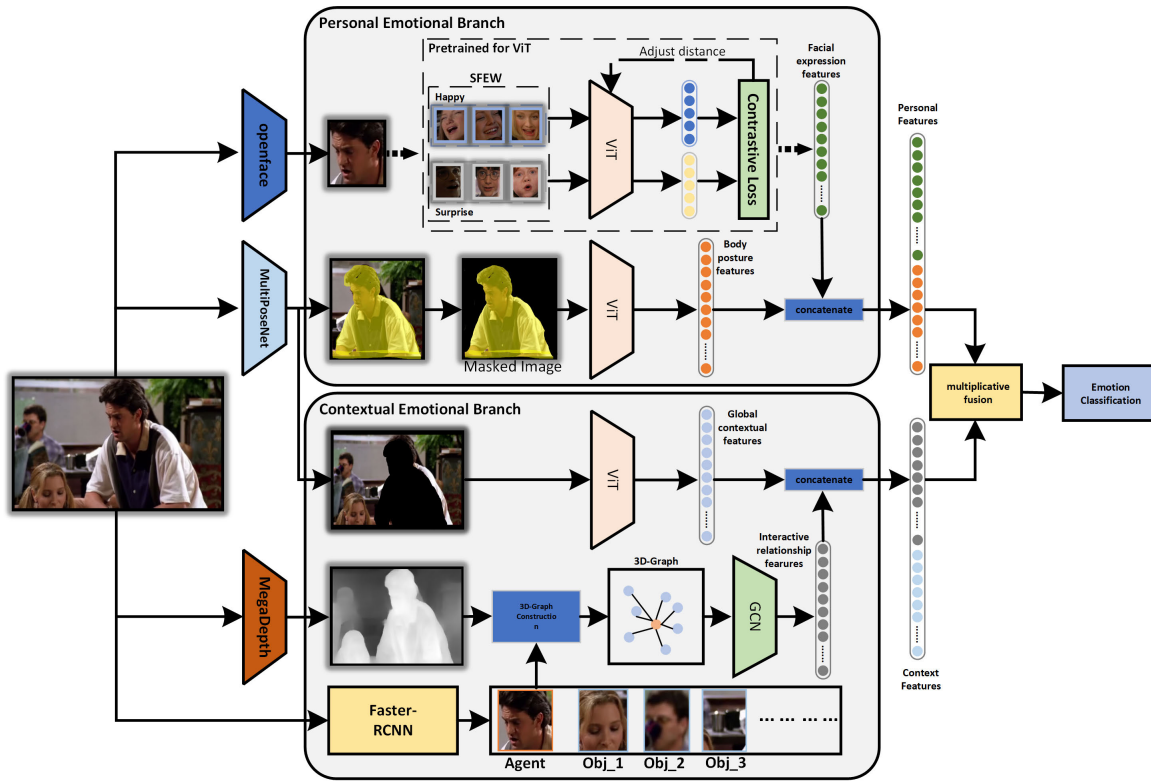


Fig. 2. Overview of TDRCer. TDRCer has two branches which are PEB and CEB. Each branch is divided into two subbranches. Two subbranches of PEB are used to extract facial expressions and body posture features, respectively. The two subbranches of CEB are used to extract background context features and object interaction features in the context, respectively. Multiplication fusion strategy is applied to fuse the features of PEB and CEB.

posited that the proximity of individuals to objects in a scene influences social behavior. Building on this premise, Mittal et al. [27] proposed EmotiCon, utilizing the depth map to illustrate the influence of social distance on people's emotions in the emotion recognition process. Wu et al. [16] built a scene graph based on the labels of objects and integrated facial expressions, body postures, and background features into emotion recognition. Zhang et al. [42] constructed emotion graphs, in which nodes were created in the scene by region proposal network and a CNN was utilized to extract features of the global scene context. Gao et al. [43] proposed GRERN to construct videos into graphs, the GCN was applied to extract temporal and spatial features, and then, the GRU was utilized to remove redundant features for emotion recognition.

Most previous studies represent emotional features at the image level, without fully modeling interactions between individuals and objects in the scene, which limits the usage of environmental emotional cues. To address this, we propose a method that incorporates gaze angles and depth maps to capture emotional features in human interactions, enhancing the performance of CAER models.

III. PROPOSED METHOD

To address the aforementioned challenges, TDRCer is proposed. The overview of TDRCer is illustrated in Fig. 2. First, two ViTs are applied to PEB to extract personal-related emotional features. A ViT which pretrained by contrastive learning with a novel contrastive loss is employed to enhance the robustness of facial expression features. Simultaneously,

another ViT is applied to process body contour images to extract body posture features. Second, CEB is constructed to extract context-related features. The agent-masked images are fed into a ViT for global context extraction. Meanwhile, a 3DVG, constructed based on gaze angles and depth maps, models interactions between agents and objects. A GCN is introduced to extract emotional features from 3DVG. Finally, multiplicative fusion integrates personal and context-related emotional features, emphasizing relevant information while filtering out irrelevant features.

A. Feature Extraction for Facial Expression and Body Posture by PEB

Personal emotional information prominently encompasses two components in visual emotion recognition: facial expressions and body postures. The following is an overview of PEB.

First, a ViT is pretrained using contrastive learning with a novel loss function that combines Euclidean distance and cosine similarity to enhance the robustness of facial expression features. Second, another ViT is used for body posture feature extraction, employing a semantic segmentation mask to isolate the human body from the background. Finally, facial expression and body posture features are integrated to form personal emotional features.

1) *Facial Expression Feature Extraction Based on Contrastive Learning*: In large-scale scenes, variations in illumination and image quality can hinder the extraction of critical features. To address these challenges, we design a novel loss function that combines Euclidean distance with cosine

similarity, applying it to contrastive learning to pretrain a ViT for facial expression feature extraction. The construction details of this novel loss function are given as follows.

Let I_1 and I_2 represent two images. The pair of images, denoted as $\langle I_1, I_2 \rangle$, constitutes the input for the ViT. $f(\cdot)$ is applied to signify the feature extraction process using ViT. Accordingly, the pair of features transformed by ViT is $\langle f(I_1), f(I_2) \rangle$. The Euclidean distance between them is calculated as follows:

$$\begin{aligned} \text{dis}_{\text{Euclid}}(I_1, I_2) &= D_{\text{Euclid}}(f(I_1), f(I_2)) \\ &= \|f(I_1) - f(I_2)\|_2^2 \end{aligned} \quad (1)$$

where $f(I_1)$ and $f(I_2)$ represent the output features extracted by ViT of images I_1 and I_2 , respectively.

The loss function proposed by Meng et al. [44], formulated based on the Euclidean distance, is defined as follows:

$$\mathcal{L}_{\text{contrastive}}^{\text{Euclid}} = \sum_{i=1}^M L_1(y_i, f(I_{i,1}), f(I_{i,1})) \quad (2)$$

$$\begin{aligned} L_1(y_i, f(I_{i,1}), f(I_{i,1})) &= y_i \times D_{\text{Euclid}}(f(I_{i,1}), f(I_{i,2})) \\ &+ (1 - y_i) \times \max(0, \alpha - D_{\text{Euclid}}(f(I_{i,1}), f(I_{i,2}))) \end{aligned} \quad (3)$$

$$y_i = \begin{cases} 1, & \text{if } \text{cls}(I_{i,1}) = \text{cls}(I_{i,2}) \\ 0, & \text{if } \text{cls}(I_{i,1}) \neq \text{cls}(I_{i,2}) \end{cases} \quad (4)$$

where $\mathcal{L}_{\text{contrastive}}^{\text{Euclid}}$ denotes the loss function formulated based on the Euclidean distance, while M represents the total number of image pairs (e.g., $\langle I_1, I_2 \rangle$). A triplet is denoted as $(y_i, f(I_{i,1}), f(I_{i,1}))$, where $f(I_{i,1}), f(I_{i,1})$ represents the extracted feature vectors for the first and second images of the i th image pair. y_i indicates whether $I_{i,1}$ and $I_{i,2}$ belong to the same class. The mathematical formula is expressed in (4), where $y_i = 1$ when $I_{i,1}$ and $I_{i,2}$ are samples of the same class; otherwise, $y_i = 0$. $\alpha > 0$ represents a threshold. Optimizing $\mathcal{L}_{\text{contrastive}}^{\text{Euclid}}$ enables the reduction of distances between similar samples in the feature space while simultaneously pushing apart the distances between samples from different classes.

Solely depending on the Euclidean distance may result in the blending of samples on the class boundary. To address this limitation, a cosine similarity loss function is introduced to provide further constraints in feature space. Similarly, let I_1 and I_2 denote two images and $\langle I_1, I_2 \rangle$ represent the image pair composed of I_1 and I_2 . The feature pair transformed by ViT is $\langle f(I_1), f(I_2) \rangle$. The cosine value of these two vectors is expressed in the following equation:

$$\begin{aligned} \text{dis}_{\text{cosine}}(I_1, I_2) &= S_{\text{cosine}}(f(I_1), f(I_2)) \\ &= \frac{\sum_{i=1}^n (x_i \times y_i)}{\sqrt{\sum_{i=1}^n x_i^2} \times \sqrt{\sum_{i=1}^n y_i^2}} \end{aligned} \quad (5)$$

where S_{cosine} denotes the cosine similarity function; the range of $\text{dis}_{\text{cosine}}(I_1, I_2)$ is $[-1, 1]$.

Following that, the cosine similarity loss function is defined in (6)

$$\begin{aligned} \mathcal{L}_{\text{contrastive}}^{\text{cosine}} &= \sum_{i=1}^M L_{\text{cosine}}(y_i, f(i_1), f(i_2)) \\ L_{\text{cosine}}(y_i, f(I_{i,1}), f(I_{i,2})) & \end{aligned} \quad (6)$$

$$\begin{aligned} &= -y_i \times \beta \times S_{\text{cosine}}(f(I_{i,1}), f(I_{i,2})) \\ &+ (1 - y_i) \times \beta \\ &\times \max(0, S_{\text{cosine}}(f(I_{i,1}), f(I_{i,2}))) \end{aligned} \quad (7)$$

where $\mathcal{L}_{\text{contrastive}}^{\text{cosine}}$ denotes the loss function formulated based on the cosine similarity between $f(I_{i,1})$ and $f(I_{i,2})$. M denotes the number of image pairs, and β signifies a hyperparameter. Analogous to α in (3), when $y_i = 0$ and $\cos(f(I_{i,1}), f(I_{i,2})) > 0$, it indicates a small angle between vectors of different categories, implying a high similarity. Consequently, there is a need to increase the angle between these two samples.

Two loss functions, based on the Euclidean distance and cosine similarity, optimize interclass and intraclass distances in both absolute distance and angles, respectively. The fused loss function is expressed as follows:

$$L_{\text{contrastive}} = \lambda_1 \times \mathcal{L}_{\text{contrastive}}^{\text{Euclid}} + \lambda_2 \times \mathcal{L}_{\text{contrastive}}^{\text{cosine}} \quad (8)$$

$$\lambda_1 + \lambda_2 = 1 \quad (9)$$

where λ_1 and λ_2 represent the weights when the two loss functions are fused.

The pretraining process involves the incorporation of the SFEW [49] dataset, which is similarly collected in the wild for facial expression recognition. The pseudocode for pretraining ViT on the SFEW [49] dataset is presented in Algorithm 1.

The ViT pretrained by contrastive learning is employed for facial expression feature extraction. While retaining the parameters of the feature extraction network, a fully connected layer is appended to the pretrained ViT. Subsequently, the pretrained ViT will be fine-tuned on the CAER dataset.

Algorithm 1 ViT Pretraining for Facial Expression Feature Extraction Based on Contrastive Learning

Input: SFEW [49] Dataset D , ViT parameters $\theta_1^{\text{encoder}}$;

Output: Pretrained ViT encoder parameters $\theta_1^{\text{encoder}'}$.

- 1: Initialize the parameters randomly: $\theta_1^{\text{encoder}}$;
 - 2: Initialize a triple set: D_{triple} ;
 - 3: **for** $i = 1 \rightarrow M$ **do**
 - 4: Select randomly image pairs from D : I_1, I_2 ;
 - 5: **if** $I_1, I_2 \in \text{Same Category}$ **then**
 - 6: $\text{construct_triple} \leftarrow \langle 1, I_1, I_2 \rangle$;
 - 7: **else**
 - 8: $\text{construct_triple} \leftarrow \langle 0, I_1, I_2 \rangle$;
 - 9: **end if**
 - 10: $D_{\text{triple}}.\text{append}(\text{construct_triple})$;
 - 11: **end for**
 - 12: **for** $i = 1 \rightarrow N_{\text{epoche}}$ **do**
 - 13: **for** $i = 1 \rightarrow N_{D_{\text{triple}}}$ **do**
 - 14: $\langle \text{flag}, I_1, I_2 \rangle \leftarrow D_{\text{triple}}[i]$;
 - 15: $F_1^i \leftarrow \text{ViT}(I_1, \theta_1^{\text{encoder}})$;
 - 16: $F_2^i \leftarrow \text{ViT}(I_2, \theta_1^{\text{encoder}})$;
 - 17: Calculate the loss by (8);
 - 18: Update ViT encoder parameters $\theta_1^{\text{encoder}}$;
 - 19: **end for**
 - 20: **end for**
 - 21: **return** $\theta_1^{\text{encoder}'} \leftarrow \theta_1^{\text{encoder}}$.
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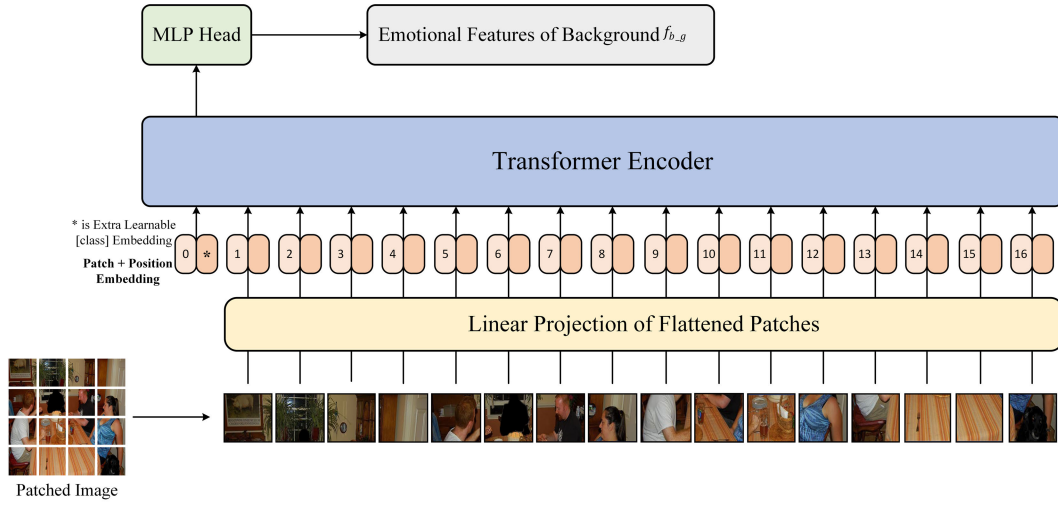


Fig. 3. Global contextual emotional features extraction based on ViT. The patched image is first transformed into feature vectors by linear projection. Each feature vector is embedded with an additional position to preserve the relative position between patches. The encoder is then used to extract the features of the patch embedding. Finally, global contextual emotion features are obtained by the MLP Head.

2) *Body Posture Feature Extraction*: The human body posture in an image can be approximated and represented by body contouring, which is extracted using MultiPoseNet [46]. To facilitate the acquisition of human pose features, we employ object detection and semantic segmentation on the image, as illustrated in Fig. 2.

Denote I_{seg_p} as the image after semantic segmentation and I as the original image in the datasets

$$\text{pixel}_{seg_p} = \begin{cases} \text{pixel}, & \text{if pixel} \in \text{body} \\ 0, & \text{if pixel} \notin \text{body} \end{cases} \quad (10)$$

where pixel_{seg_p} represents each pixel in I_{seg_p} and pixel denotes each pixel in I . If a pixel in I_{seg_p} belongs to the human body, it is set to pixel ; otherwise, it is set to 0. Executing this operation for all pixels in the image results in I_{seg_p} .

Subsequently, I_{seg_p} is fed into ViT for extracting emotional features related to the human body pose.

3) *Feature Fusion in PEB*: As illustrated by PEB in Fig. 2, we employ feature-level fusion to integrate emotional features from both facial expression and body posture.

Denote v_{peb} as the emotional feature vector extracted in the PEB. The construction process for v_{peb} is outlined in the following equation:

$$v_{peb} = \text{concat}(v_{fe}, v_{bp}) \quad (11)$$

where $\text{concat}()$ denotes a function applied to concatenate two vectors. v_{fe} and v_{bp} represent the extracted facial expression features and body posture features, respectively.

B. Feature Extraction for Scene Context by CEB

Scene context information encompasses two aspects: the global context (background) and the interactive relationships between agents and surrounding objects. A detailed description of the construction for CEB is given as follows.

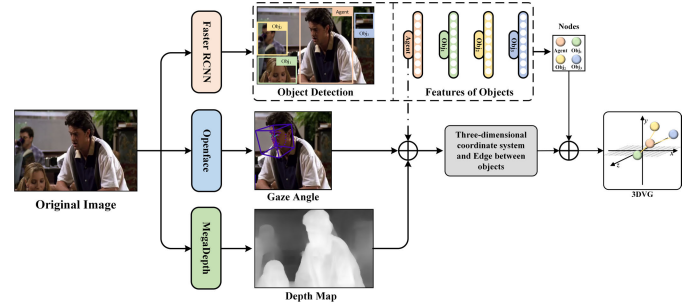


Fig. 4. Construction of 3DVG. In 3DVG, nodes represent the objects detected within the image, with each node's feature defined by the corresponding object's feature vector. The relationships between nodes are established based on depth maps, gaze angles, and the relative positions of the objects.

1) *Global Contextual Feature Extraction*: The personal information (including face and body) in image I needs to be masked to compel the network to focus on valuable global contextual details, as shown in Fig. 3. In PEB, we obtained the human body contour I_{seg_p} to represent the body posture. In this branch, the human body will be occluded to retain the background features in the image. The pixels in I_{seg_p} are initially “inverted” to generate a mask for the image and computed as follows:

$$\text{pixel}_{mask_bg} = \begin{cases} 1, & \text{if } \text{pixel}_{seg_p} = 0 \\ 0, & \text{if } \text{pixel}_{seg_p} \neq 0 \end{cases} \quad (12)$$

where pixel_{mask_bg} refers to the pixels in the mask, while pixel_{seg_p} represents the pixels in the body contour image I_{seg_p} . The operation in the creation of a mask is denoted as I_{mask_bg} .

Subsequently, I_{mask_bg} is employed to mask the body contour in the source image I , retaining the background information

$$I_{bg} = I \odot I_{mask_bg} \quad (13)$$

where $\odot$ denotes pixelwise multiplication between the source image I and the background mask I_{mask_bg} . By applying the mentioned masking operation to the source image, we acquire

an image I_{bg} that exclusively encapsulates background information.

Finally, similar to PEB, we use a ViT as the backbone to extract emotional features from I_{bg} , yielding the features f_{bg} .

2) *Feature Extraction for Interactive Relationship Among Objects*: To model the interactive relationship between objects and objects in the scene and capture their emotional cues effectively, a three-dimensional view method for representing human-object relationship is established to construct 3DVG, as illustrated in Fig. 4.

a) *Definition of nodes in 3DVG*: First, Faster-RCNN [48] is employed for object detection in image I to identify the position of each object, denoted as obj_i . The coordinates of the upper left corner of the bounding box of obj_i are $(x_1^{obj_i}, y_1^{obj_i})$, and the coordinates of the lower right corner are $(x_2^{obj_i}, y_2^{obj_i})$ post-object detection. The position of obj_i is determined by setting the coordinates of the central point of the bounding box. The calculation is given as follows:

$$pst_{obj_i} = \left(\frac{x_1^{obj_i} + x_2^{obj_i}}{2}, \frac{y_1^{obj_i} + y_2^{obj_i}}{2} \right) \quad (14)$$

where pst_{obj_i} represents the positional coordinates of obj_i in source image I .

Subsequently, OpenFace [45] is employed to extract the agents' gaze angle from the image. The gaze angle (ga) extracted using OpenFace [45] comprises two components: ga_x , representing the left-right gaze angle, and ga_y , representing the up-down gaze angle. Both variables are expressed in radians.

The three-dimensional coordinate representation rule is defined on the image, where the width of the image corresponds to the positive direction of the x -axis, the opposite height of the image corresponds to the positive direction of the y -axis, and the direction orthogonal to the two axes is the z -axis. 3DVG is then constructed based on the three coordinate axes.

The first node identified in 3DVG is denoted as its central node N_{agent} , representing the human node for the target of emotion recognition. The coordinates of the central position of the agent's face, pst_{agent} , are employed as the coordinates of the central node, while the features X_{agent} extracted by Faster-RCNN [48] from the agent region serve as the features of the central node. Analogously, each object identified in the image serves as a node in 3DVG, denoted as N_{obj_i} .

During the object detection, the convolutional module in Faster-RCNN [48] produces features X_{obj_i} for each candidate region, and the features will be utilized as the node features for obj_i in 3DVG. Subsequently, the relationship between the agent and each obj_i is modeled.

b) *Definition of edges in 3DVG*: As previously discussed, the gaze angle ga obtained by OpenFace [45] comprises two elements: $ga = (ga_x, ga_y)$. Here, ga_x is considered positive when the agent's viewpoint is inclined toward the positive direction of the x -axis in the xOz plane, and it measures 0° when aligned with the z -axis direction. Likewise, ga_y is considered positive when the agent's viewpoint is inclined toward the positive direction of the y -axis in the yOz plane,

implying a downward gaze, and it registers 0° when aligned with the z -axis direction. Both angles represent the deviation of the line of sight from the positive direction of the z -axis.

Moreover, the vector a_{ga} , representing the gaze direction, is determined by the angles (ga_x, ga_y) , originating from the coordinates $(0, 0, 0)$. The projection of a_{ga} onto the xOz plane is represented by the unit vector u , and the projection onto the yOz plane is represented by the vector v . The coordinates of two vectors are computed from ga_x, ga_y

$$u = (\sin ga_x, \cos ga_x) \quad (15)$$

$$v_y = \cos ga_x \times \tan ga_y \quad (16)$$

$$v = (0, \cos(ga_x)) \quad (17)$$

where v_y represents the y -axis coordinate of the vector v . The vector a_{ga} , indicating the gaze direction, is derived from the two projection vectors as follows:

$$\begin{aligned} a_{ga} &= (\sin ga_x, v_y, \cos ga_x) \\ &= (\sin(ga_x), \cos ga_x \times \tan ga_y, \cos(ga_x)). \end{aligned} \quad (18)$$

Subsequently, three-dimensional orientation vectors from the agent to other objects in the image are further constructed. MegaDepth [47] is employed to produce a depth map, illustrating the relative distance of each pixel in the image from the camera; the lower the pixel value in the depth map, the farther the pixel is from the camera.

Let I_{depth} represent the depth map of the original image I and consider the bounding box of each object, denoted as obj , obtained by Faster-RCNN [48]. The Euclidean distance from the centroid coordinates pst_{obj_i} of the object obj_i , as mentioned previous, to the central coordinates pst_{agent} of the agent is calculated, designated as dis_{ao} .

Then, the three-dimensional coordinates of obj_i in the coordinate system are derived from I_{depth} . The x -axis and y -axis coordinates of obj_i are aligned with the x -axis and y -axis coordinates of pst_{obj_i} , respectively. In addition, I_{depth} indicates the distance of a pixel from the camera by the value of every pixel. The pixel of the center point of obj_i in I_{depth} is denoted as $p_{obj_i}^{dep}$, and then, the representation of z for obj_i in the coordinate system is expressed in the following equation:

$$z_{obj_i} = p_{obj_i}^{dep} - p_{agent}^{dep} \quad (19)$$

where p_{agent}^{dep} represents the pixel value of the agent's center in I_{depth} . z_{obj_i} is positive when obj_i is in front of the agent; otherwise, it is negative. This assignment aligns with the construction process of the gaze direction vector a_{ga} . For computational convenience, we designate the plane where the agent coordinates are located as the xOy plane, implying that the z -axis value is 0; in other words, the coordinates of the agent are $(x_{agent}, y_{agent}, 0)$. The Euclidean distance dis_i between the agent and obj_i in three-dimensional coordinates is then calculated as follows:

$$\begin{aligned} dis_i &= \sqrt{(x_{agent} - x_{obj_i})^2 + (y_{agent} - y_{obj_i})^2 + (0 - z_{obj_i})^2} \\ &= \sqrt{dis_{ao}^2 + z_{obj_i}^2} \end{aligned} \quad (20)$$

where dis_{ao} represents the Euclidean distance between the agent and obj_i in a 2-D coordinate system. Obtaining the above

equation is straightforward using the Pythagorean theorem. The vector pointing from the agent to obj_i can be expressed as follows:

$$\begin{aligned} \mathbf{v}_i &= (x_{obj_i}, y_{obj_i}, z_{obj_i}) - (x_{agent}, y_{agent}, 0) \\ &= \left(\frac{x_1^{obj_i} + x_2^{obj_i}}{2} - x_{agent}, \frac{y_1^{obj_i} + y_2^{obj_i}}{2} \right. \\ &\quad \left. - y_{agent}, p_{obj_i}^{dep} - p_{agent}^{dep} \right). \end{aligned} \quad (21)$$

Subsequently, the relationships (i.e., edges) between N_{agent} and N_{obj_i} are represented by utilizing the gaze direction vectors $\mathbf{a}_{ga}$, dis_i , and $\mathbf{v}_i$. The smaller angle between $\mathbf{a}_{ga}$ and $\mathbf{v}_i$ represents the stronger interactive relationship. In other words, an object in the direction of the agent's gaze has a significant impact on emotions. Considering the monotonicity of the cosine function in the range $[0, \pi]$, the weights w_i between N_{agent} and N_{obj_i} are calculated by the following equations:

$$\cos \theta = \frac{\mathbf{a}_{ga} \cdot \mathbf{v}_i}{|\mathbf{a}_{ga}| |\mathbf{v}_i|} \quad (22)$$

$$w_i = \cos \theta \times \frac{1}{dis_i} \times \frac{\sum_{j=1, j \neq i}^n dis_j}{\sum_{j=1}^n dis_j} \quad (23)$$

where $\cos \theta$ represents the cosine of the angle between the vectors $\mathbf{a}_{ga}$ and $\mathbf{v}_i$. The term $(\sum_{j=1, j \neq i}^n dis_j / \sum_{j=1}^n dis_j)$ is a weight assignment factor designed to allocate larger weights to objects closer to the agent during graph construction. Here, n denotes the total number of objects in the image.

c) *Construction of 3DVG*: 3DVG is constructed based on the aforementioned nodes and edges, and the GCN is employed to extract emotional features from 3DVG. Algorithm 2 describes the 3DVG construction process in detail.

Let $G = \{V, E, X\}$, $V = \{N_{agent}, N_{obj_1}, \dots, N_{obj_n}\}$ denote the node set in G , comprising the detected objects in the image. $E = \{e_1, e_2, \dots, e_m\}$ represents the edge set in G , consisting of w_i . $X \in \mathbb{R}^{(n+1) \times l}$ signifies the feature vector matrix of nodes, encompassing $n+1$ nodes, with the length of feature in each node being l . The constructed graph structure $G = \{V, E, X\}$ for each image is inputted into a GCN to acquire the emotional features f_{c_r} .

3) *Feature Fusion in CEB*: To obtain robust scene context information, f_{bg} and f_{c_r} are fused. Concatenating the two vectors, we can obtain the feature vector f_c for the scene context by the following equation:

$$f_{_c} = \text{concat}(f_{bg}, f_{c_r}) \quad (24)$$

where $\text{concat}()$ is the concatenation function, which means that the two vectors are stitched together. f_{bg} and f_{c_r} are the features of the global contextual feature and the local contextual feature.

C. Loss Function

1) *Multiplicative Fusion*: As shown in previous research [27], multiplicative fusion serves to emphasize important features from multiple modalities while suppressing redundant ones. To enhance the robustness of the features

Algorithm 2 Construction of 3DVG

Input: Original image I ;

Output: Constructed 3DVG $G = \{V, E, X\}$;

- 1: Get objects by Faster-RCNN: $(Objs, Pos', Feats) \leftarrow R(I)$; # $R()$ represents the process of Faster-RCNN
- 2: $Objs \leftarrow \{obj_0, obj_1, \dots, obj_n\}$;
- 3: $Pos' \leftarrow \{pos_0, pos_1, \dots, pos_n\}$;
- 4: $Feats \leftarrow \{feat_0, feat_1, \dots, feat_n\}$;
- 5: Construct the node set of 3DVG by $Objs$:
- 6: $V \leftarrow \{N_{agent}, N_{obj_1}, N_{obj_2}, \dots, N_{obj_n}\}$
- 7: $\quad = \{obj_0, obj_1, \dots, obj_n\}$;
- 8: Assign the features of Nodes in 3DVG:
- 9: $X \leftarrow \{x_{agent}, x_1, \dots, x_n\}$;
- 10: Get the gaze angle of agents in V by OpenFace:
- 11: $ga \leftarrow (ga_x, ga_y) = \text{Openface}(N_{agent})$;
- 12: Get the depth map of I by MegaDepth:
- 13: $I_{depth} \leftarrow \text{MegaDepth}(I)$;
- 14: A 3-dimensional coordinate system is constructed by ga and I_{depth} , and the distance between each obj and $agent$ in the 3-dimensional coordinate system is calculated by (17)~(21);
- 15: Construct the edge set of 3DVG:
- 16: $E \leftarrow \{w_1, w_2, \dots, w_n\}$ # w_i is calculated by (23);
- 17: **return** $G \leftarrow \{V, E, X\}$.

extracted by our model, the multiplicative fusion [27] is applied to efficiently integrate the features extracted by PEB and CEB. The formula for multiplicative fusion is presented as follows [27]:

$$\mathcal{L}_{\text{multiplicative}} = - \sum_{i=1}^n (p_i^e)^{\frac{1}{n-1}} \log p_i^e \quad (25)$$

where n is the total number of modalities being considered and p_i^e is the prediction for emotion class e , given by the network for the i th modality. In this article, the number of modalities n is 2, which represents the features extracted by PEB and CEB, respectively.

2) *Loss Function of the Proposed Method*: The loss function applied to model training is shown as follows:

$$\mathcal{L} = \mathcal{L}_{\text{multiplicative}} + \mathcal{L}_{\text{classification}} \quad (26)$$

where $\mathcal{L}_{\text{classification}}$ is the loss for emotion classification, which is different for different datasets.

IV. EXPERIMENTS AND DISCUSSION

A. Datasets

We evaluate the proposed method on two datasets, namely, EMOTIONS in context (EMOTIC) [17] and CAER-S [38] datasets. The details of the two datasets are provided in the following.

The EMOTIC [17] dataset comprises images collected in the wild, annotated with apparent emotions of individuals. This dataset encompasses 23 571 images and includes annotations for 34 320 individuals. EMOTIC is labeled with both discrete and continuous sentiment. The discrete classification includes 26 emotion categories.

The CAER [38] dataset is a comprehensive benchmark that encompasses both images and videos. It comprises over 13 000 annotated videos and 70 000 annotated static images. The annotations include seven emotion categories [38], which are *Anger*, *Disgust*, *Fear*, *Happy*, *Neutral*, *Sad*, and *Surprise*. In this article, we utilize static images from CAER to assess the proposed method, referring to the subset of CAER with static images as CAER-S.

B. Evaluation Metrics

1) *Multiclassification Metrics*: The accuracy of emotion classification (i.e., CAER-S [38]) intuitively reflects the precision of the model in identifying samples. Let TP_i represent the number of instances correctly predicted with the real label, FN_i represent the number of errors where the model predicts the real label i as other labels, $F_i P_j$ represent the number of instances where the real label i is incorrectly predicted as label j , and TM represent the total number of samples. The mathematical expressions for the aforementioned classification indicators are shown in the following equations:

$$FN_i = \sum_{j=1, j \neq i}^n F_i P_j \quad (27)$$

$$Acc = \frac{\sum_{i=1}^n TP_i}{\sum_{i=1}^n TP_i + FN_i} \quad (28)$$

where n represents the number of emotion categories in the datasets and Acc denotes the calculated accuracy of the model on the dataset.

2) *Multilabel Classification Metrics*: As the EMOTIC [17] dataset assigns multiple labels to each sample, the mean average precision (mAP) is employed to assess the model's performance. The mathematical expression is shown in the following equation:

$$mAP = \frac{\sum_{i=1}^N AveP(i)}{N} \quad (29)$$

where N is the quantity of samples and $AveP(i)$ is the average predictive accuracy of the i th test sample. $AveP(i)$ is calculated by the following equation:

$$AveP(i) = \frac{\sum_{k=1}^n (P_i(k) \times correct_i(k))}{m_i} \quad (30)$$

where n represents the number of candidate types. m_i represents the number of categories actually labeled to the i th sample. When calculating $AveP(i)$, the scores of each candidate are first sorted from high to low and then judged whether it is the correct label. $P_i(k)$ is the accuracy of the i th test sample up to the k th candidate type. $correct_i(k)$ is an indicator function, and the value is 1 only when the k th candidate type is the correct label; otherwise, it is 0.

mAP is chosen as the evaluation metric for context emotion recognition models due to its effectiveness in handling multi-class scenarios, managing imbalanced datasets, accounting for emotion ranking, and providing a comprehensive assessment of both false positives and false negatives. Its robustness to varying confidence levels further enhances its suitability for emotion recognition. Compared to metrics such as accuracy,

recall, and precision, mAP offers a more nuanced and versatile evaluation, facilitating a holistic understanding of model performance. In addition, mAP is widely used for comparing the performance of different context emotion recognition methods.

C. Implementation Details

In this section, the implementation details and experimental settings of our method are introduced. The model was trained on two NVIDIA RTX A6000 GPUs, each equipped with 48 GB of RAM.

In the proposed method, we apply two backbones, ViT and GCN. ViT is used to extract image features, including facial expression, body posture, and global context features. In each branch, the patch size of ViT is set to 16×16 , and the size of the input image is resized to 224×224 . The depth of ViT is set to 12, and the head of the multihead attention mechanism is set to 8. A GCN is applied to extract emotional features from 3DVG. The number of layers of GCN is set to 4, and ReLU is applied to the activation function.

Before training the whole model, we pretrained the ViT for the feature extraction of facial expressions. The static facial expression dataset in film named SFEW [49] is employed in the process of pretraining. In the experiments, the parameters λ_1 and λ_2 are equally set to 0.5, respectively. The values of λ_1 and λ_2 will be discussed in the section of *Ablation study*. Besides, the parameters α and β in (3) and (7) are equally set to 5, experientially.

The model is implemented by applying the PyTorch library. The learning rate is initialized to 0.001 during the training of the model. The Adam optimizer is applied and has a momentum of 0.9 with a weight decay of 0.05. The model is trained for a total of 150 epochs with a batch size of 64.

Our proposed model incorporates three ViT and a four-layer GCN. Subsequently, a two-layer classifier is employed for emotion classification. In the initial layer of the classifier, 1024-D vectors serve as inputs, producing 512-D outputs. The second layer accepts inputs of 512 dimensions and yields outputs corresponding to the number of emotion categories. In accordance with the specified ViT model parameters, each ViT model contains 76.65 M parameters, while the GCN comprises 0.79 M parameters and the classifier involves 0.52 M parameters (as exemplified using the CAER-S [38] dataset). Consequently, the cumulative number of parameters in the overall model amounts to 231.11 M. Furthermore, we computed the floating point operations (FLOPs) for our proposed method. The FLOPs utilized by each ViT amount to 15.06 G, while the FLOPs employed by the GCN stand at 0.16 G. The classifier consumes 1.01 M FLOPs. Consequently, the aggregate computational capacity utilized by the model approximates 45.34 G.

D. Experimental Results and Analysis

We assess the proposed method using the CAER-S [38] dataset and the EMOTIC [17] dataset. The corresponding experimental results are presented in Tables I–III.

TABLE I
COMPARISON WITH OTHER METHODS ON THE CAER-S [38] DATASET

Method	Accuracy
CAER-Net-S [38]	73.51%
GRERN [43]	81.31%
Jaiswal <i>et al.</i> [29]	80.79%
Li <i>et al.</i> [28]	84.82%
TDRCer (ours)	89.90%

TABLE II
COMPARISON OF EACH CATEGORY WITH OTHER METHODS
ON THE CAER-S [38] DATASET

Emotion Categories	GRERN [43]	Li <i>et al.</i> [28]	Jaiswal <i>et al.</i> [29]	TDRCer (ours)
Anger	82%	88%	88%	91%
Disgust	90%	91%	93%	97%
Fear	91%	94%	95%	99%
Happy	76%	82%	69%	83%
Neutral	65%	69%	60%	77%
Sad	85%	92%	80%	93%
Surprise	81%	80%	79%	91%
Acc	81.31%	84.82%	80.79%	89.90%

1) *Results on CAER-S Dataset*: The experimental results of our method on CAER-S [38] dataset are illustrated in Tables I and II and Fig. 5. Table I provides a comparison of our method with other methods in terms of accuracy, while Table II illustrates the performance of our method in each category compared to other methods. Fig. 5 demonstrates the confusion matrix of our method. We compare our method with CAER-Net-S [38], GRERN [43], and the method proposed by Li *et al.* [28] and Jaiswal *et al.* [29].

In Table I, it can be observed that our method outperforms most other methods solely based on visual cues. CAER-Net-S [38] pioneered the exploration of CAER on the CAER-S [38] dataset, and it extracts emotional features from facial expressions and global contextual information. However, it does not account for the impact of body posture and the relationship between contextual objects on emotional information. Our model achieves a 16.39% accuracy improvement over CAER-Net-S. GRERN [43] constructs a semantic graph by semantic information of objects in the scene and employs a GCN to extract emotional features in the semantic graph. GRERN [43] achieves an accuracy of 81.31%. In comparison, our method shows an improvement of 8.59% in accuracy. Jaiswal *et al.* [29] extract the features from facial expression and scene context, attention mechanism is utilized to improve the relevant prediction, and the method achieves an accuracy of 80.97%. Li *et al.* [28] utilize the body information and the objects in the scene to recognition the emotion and achieve an accuracy of 84.82%; our method outperforms it 5.08%. According to the results presented in Table I, our method exhibits an enhanced accuracy in CAER.

Table II presents the accuracy of our method and other methods on each emotion category. It can be observed that our method significantly outperforms other methods. The accuracy of the *Neutral* and *Surprise* is notably higher than

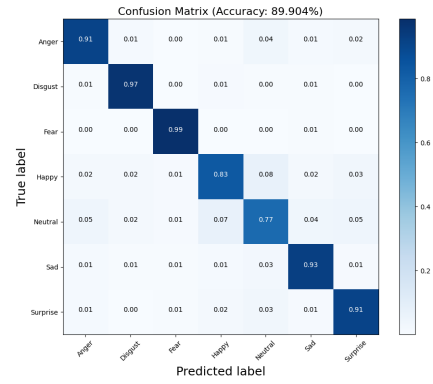


Fig. 5. Confusion matrix of TDRCer on the CAER-S [38] dataset.

TABLE III
COMPARISON WITH OTHER METHODS ON THE EMOTIC [17] DATASET

Emotion Categories	Zhang <i>et al.</i> [42]	EmotiCon [27]	Kosti <i>et al.</i> [19]	Li <i>et al.</i> [28]	TDRCer (ours)
Affection	46.89%	45.23%	27.85%	37.93%	43.23%
Anger	10.87%	15.46%	9.49%	13.73%	16.15%
Annoyance	11.27%	21.92%	14.06%	20.87%	20.22%
Anticipation	62.64%	72.12%	58.64%	61.08%	68.34%
Aversion	5.93%	17.81%	7.48%	9.61%	15.22%
Confidence	72.49%	68.85%	78.35%	80.08%	64.35%
Disapproval	11.28%	19.82%	14.97%	21.54%	30.33%
Disconnection	26.91%	43.12%	21.32%	28.32%	44.35%
Disquietment	16.94%	18.73%	16.89%	22.57%	27.12%
Doubt/Confusion	18.68%	35.12%	29.63%	33.50%	32.11%
Embarrassment	1.94%	14.37%	3.18%	4.16%	16.38%
Engagement	88.56%	91.12%	87.53%	88.12%	88.27%
Esteem	13.33%	23.62%	17.73%	20.50%	24.62%
Excitement	71.89%	83.26%	77.16%	80.11%	82.66%
Fatigue	13.26%	16.23%	9.70%	17.51%	19.87%
Fear	4.21%	23.65%	14.14%	15.56%	27.21%
Happiness	73.26%	74.71%	58.26%	76.01%	72.11%
Pain	6.52%	13.21%	08.94%	14.56%	15.19%
Peace	32.85%	34.27%	21.56%	26.76%	36.27%
Pleasure	57.46%	65.53%	45.46%	55.64%	64.21%
Sadness	25.42%	23.41%	19.66%	30.80%	25.22%
Sensitivity	5.99%	8.32%	9.28%	9.59%	8.24%
Suffering	23.39%	26.39%	18.84%	30.70%	25.38%
Surprise	9.02%	17.37%	18.81%	17.92%	19.37%
Sympathy	17.53%	34.28%	14.71%	15.26%	38.47%
Yearning	10.55%	14.29%	8.34%	10.11%	11.58%
mAP	28.42%	35.48%	27.38%	32.41%	36.02%

other comparative methods. In addition, Fig. 5 illustrates the confusion matrix for our model on the CAER-S [38] dataset. As depicted, our method excels in recognizing *Disgust* and *Fear* but exhibits lower accuracy in identifying *Happy* and *Neutral*. Specifically, *Neutral* tends to be confused with both *Happy* and *Sad* emotions, and *Happy* is frequently misclassified as *Neutral*. This phenomenon could be attributed to the similarity between *Neutral* and *Sad* or *Happy* in facial expression, gestures, and scene context, leading to lower accuracy of these emotions.

For an in-depth analysis of the performance of our model on the CAER-S [38] dataset, we monitored the accuracy and loss variation, as illustrated in Fig. 6(a) and (b). The figures illustrate that our model largely converges in the initial 40 epochs, after which both the accuracy and loss exhibit stability. The model achieves a consistent accuracy of 89.90%.

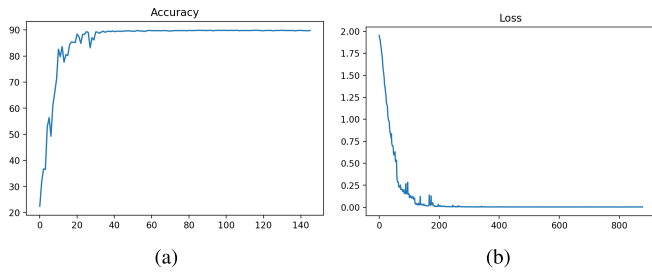


Fig. 6. Accuracy and loss of TDRCer training on the CAER-S [38] dataset. (a) Accuracy on CAER-S [38]. (b) Loss on CAER-S [38].

2) *Results on the EMOTIC Dataset:* The performance of our method is also assessed on the EMOTIC [17] dataset; Table III and Fig. 7 provide a comprehensive comparison of our method with others. The methods that we compared include EmotiCon [27], Zhang et al. [42], Kosti et al. [19], and Li et al. [28].

Among the methods in Table III, Zhang et al. [42], EmotiCon [27], Li et al. [28], and our method have considered the interaction between agents and objects in different means. Zhang et al. [42] perform emotion recognition by extracting the relationship between physical features and objects. Compared with Zhang et al. [42], our method improves mAP by 7.6%, which proves the effectiveness of our global scene context features. Kosti et al. [19] employ facial expressions and global context information, whereas our approach integrated information on body posture and interaction relationships among objects. This augmentation resulted in an 8.64% improvement in mAP. Li et al. [28] identify emotions by personal information and interaction between objects. In comparison to Li et al. [28], we introduced global scene context features, leading to a 3.61% improvement in mAP. EmotiCon [27] utilizes a depth map to represent social distance, while our method employs 3DVG to depict interactions between objects in the environment. In comparison with EmotiCon [27], our approach results in a 0.54% improvement on mAP. The experimental results indicate that the features of interaction among objects have a positive impact on emotion recognition. Our method achieves an mAP of 36.02%, surpassing the other four state-of-the-art methods. The results demonstrate the effectiveness of our method in CAER.

Meanwhile, from Table III and Fig. 7, it is evident that our method has achieved the highest levels in specific categories, such as *Disapproval*, *Disconnection*, *Embarrassment*, *Esteem*, *Fear*, *Peace*, and *Sympathy*. This also indicates the superiority of our model in CAER.

E. Ablation Study

1) *Ablation Study on CAER-S and EMOTIC Datasets:* We perform ablation experiments on CAER-S [38] and EMOTIC [17] datasets. In these experiments, we divide the model into three parts: the facial expression subbranch in PEB (PEB-FacialExp), the body pose subbranch in PEB (PEB-Pose), and CEB. Table IV illustrates the results of the ablation experiments conducted on the model.

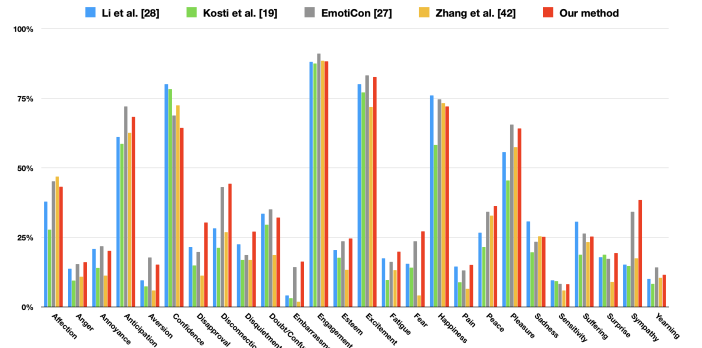


Fig. 7. Comprehensive bar chart illustrating the accuracy of our method for each emotion category on the EMOTIC [17] dataset, which can also be regarded as a bar chart representation akin to Table III.

TABLE IV
ABLATION STUDY ON CAER-S [38] AND EMOTIC [17] DATASETS.
PEB-FACIALEXP REPRESENTS THE FACIAL EXPRESSION SUBBRANCH
IN PEB. PEB-POSE REPRESENTS THE BODY
POSE SUBBRANCH IN PEB

PEB-FacialExp	PEB-Pose	CEB	CAER-S (Acc)	EMOTIC (mAP)
✓			78.08%	29.70%
✓	✓		81.25%	30.62%
✓		✓	85.77%	34.25%
✓	✓	✓	89.90%	36.02%

The model achieves an accuracy of 78.08% on the CAER-S [38] dataset and an mAP of 29.70% on the EMOTIC [17] dataset based solely on facial expression. This is attributed to the variability of facial expressions in the dataset and the occlusion of faces in some data, which led to the absence of facial expression information. Subsequently, body posture information is incorporated into facial expressions for emotion recognition, resulting in an increased accuracy of emotion recognition to 81.25% on CAER-S [38] and an mAP of 30.62% on EMOTIC [17]. Further enhancement is achieved by adding scene context information to the facial expression base, reaching a recognition accuracy of 85.77% on CAER-S [38] and an mAP of 34.25% on EMOTIC [17]. The experiments demonstrate that both body posture and contextual features can be used as supplementary features for emotion recognition.

We finally integrate facial expression, body posture, and scene context information and achieve a recognition accuracy of 89.90% on the CAER-S [38] dataset and an mAP of 36.02% on the EMOTIC [17] dataset. The inclusion of body posture features enhances accuracy by 4.23% on the CAER-S [38] dataset and an improved mAP of 1.77% on the EMOTIC [17] dataset. The three types of emotional information synergistically complement each other. In cases where a specific feature is missing or insufficient, the other two types of information serve as a foundation for emotional recognition.

2) *Discussion of λ_1 and λ_2 in Contrastive Learning:* In this section, we discuss the two parameters, denoted as λ_1 and λ_2 , where $\lambda_1 + \lambda_2 = 1$, utilized during the pretraining of ViT in Section III-A1. Fig. 8 illustrates the performance of PEB-FacialExp on CAER-S [38] and EMOTIC [17] with varying values of λ_1 .

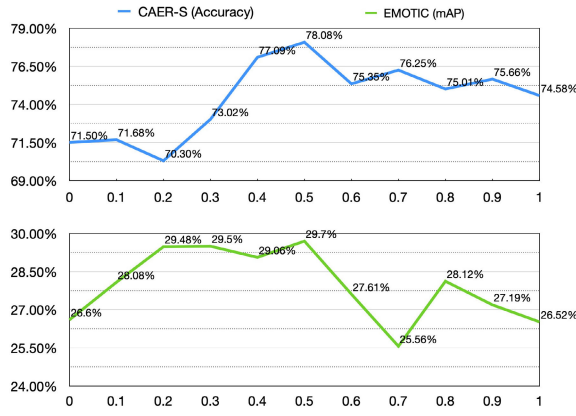


Fig. 8. Performance of PEB-FacialExp on CAER-S [38] and EMOTIC [17] datasets with varying values of λ_1 . The x-axis represents the values of λ_1 , and the y-axis depicts the accuracy of PEB-FacialExp.

In Fig. 8, it can be observed that when λ_1 is set to 0.5 during contrastive learning, PEB-FacialExp achieves the best performance on both CAER-S [38] and EMOTIC [17] datasets, with accuracy of 78.08% and mAP of 29.70%, respectively. Therefore, during contrastive learning pretraining, the values of parameters λ_1 and λ_2 are equally set to 0.5.

V. CONCLUSION AND FUTURE WORK

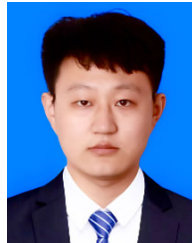
To comprehensively utilize personal- and context-related emotional features, we propose TDRCer, consisting of two main branches: PEB and CEB. PEB has two subbranches dedicated to extracting facial expression features and body posture features, and two ViTs are utilized as the backbone. A ViT is pretrained on the SFEW [49] dataset by contrastive learning with a novel loss function, and the pretrained ViT is utilized as the feature extractor of facial expressions. The human body contours obtained by semantic segmentation are employed to represent body posture, and another ViT is utilized as the feature extractor. The features of facial expression and body posture are then concatenated as the outputs of PEB. Similarly, CEB also includes two subbranches responsible for extracting contextual features from the global scene and the interactive relationship between agents and objects. Emotional features of the global scene context are extracted by ViT from images with agent-masked body contours. Meanwhile, a 3DVG is constructed to represent the interaction between agents and objects, and a GCN is applied to extract emotional features from 3DVG. Then, the features extracted by the two subbranches in CEB are also concatenated. The multiplicative fusion strategy is then applied to fuse the features from PEB and CEB for emotion recognition. The experimental results on CAER-S [38] and EMOTIC [17] datasets demonstrate the superiority of TDRCer.

In future work, we intend to explore the emotional representation between individuals and groups and propose a new framework for emotion recognition. Furthermore, we will develop an emotion recognition method aimed at diminishing the model's complexity, catering to devices with lower computing power.

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